A Three-Stage Pattern Recognition Framework for Medical Image Classification: From Manifold Learning to Massive Topology-Aware Ensembles

Emre Celil Kuru
Department of Electrical Engineering
TOBB University of Economics and Technology
Ankara, Türkiye
ekuru@etu.edu.tr

***Abstract*—We present a comprehensive three-stage pattern recognition framework evaluated across eight datasets of the MedMNIST2D benchmark collection. Stage 1 establishes baselines through UMAP manifold learning coupled with probabilistic classifiers (Gaussian Naïve Bayes, Linear and Quadratic Discriminant Analysis, and a soft-clustering Gaussian Mixture Model). Stage 2 introduces a custom hybrid neural architecture featuring a CNN front-end for efficient feature extraction, a comprehensive evaluation of seven Self-Organizing Map variants (SOM, GTM, GSOM, TASOM, Conformal, Elastic, OS-Map) and fifteen Adaptive Resonance Theory variants (ART1–3, Fuzzy ART, ARTMAP, Fuzzy ARTMAP, SFAM, Gaussian ART/ARTMAP, Fusion ART, TopoART, Hypersphere ART/ARTMAP, LAPART), a novel Tanh–GeLU activation block with negative-axis Gaussian extraction, RBF layers, and Adam–SGLD optimisation. Stage 3 implements a massive topology-aware and noise-immune pipeline encompassing Multiple Adaptive Importance Sampling, stochastic cascade augmentation with 29 methods, boundary cleaning via ADASYN and Tomek Links, Spherical Linear Interpolation-based morphological expansion, Randomized SVD dimensionality reduction, HDBSCAN-entropy coreset selection, and a 30-model Bayesian Model Averaging ensemble spanning five classifier categories with custom Nyström kernel approximations. Rigorous evaluation with accuracy, precision, recall, F1-score, and AUC across all eight datasets reveals that the Stage 2 HybridCNN-SNN achieves the best overall performance (mean accuracy 0.564, mean AUC 0.871), while aggressive topology-aware augmentation in Stage 3 requires careful per-dataset calibration to avoid information loss.**

***Index Terms*—pattern recognition, medical image classification, MedMNIST, manifold learning, spiking neural network, Bayesian model averaging, ensemble learning, dimensionality reduction**

I. Introduction

Medical image classification is a cornerstone of computer-aided diagnosis, yet the diversity of imaging modalities, class imbalance, and noise sensitivity pose significant challenges [1]. The MedMNIST2D benchmark [2] standardises evaluation across biomedical datasets spanning pathology, dermatology, ophthalmology, and organ segmentation, providing a rigorous testbed for classification algorithms.

Conventional approaches rely either on shallow classifiers applied to hand-crafted features or on deep convolutional neural networks (CNNs) trained end-to-end. However, these paradigms leave substantial room for exploration in (i) manifold-aware feature extraction, (ii) biologically inspired hybrid architectures, and (iii) topology-preserving data augmentation and ensemble strategies.

In this work, we propose a three-stage framework that progressively increases model complexity and data sophistication:

- **Stage 1** applies UMAP dimensionality reduction followed by four probabilistic classifiers, establishing interpretable baselines.
- **Stage 2** introduces a CNN-based hybrid architecture with comprehensive evaluation of 7 SOM and 15 ART variants for topology discovery, a novel Tanh–GeLU activation with Gaussian extraction, RBF layers, and Adam–SGLD optimisation.
- **Stage 3** constructs a massive noise-immune pipeline with importance sampling, cascade augmentation, boundary cleaning, SLERP expansion, SVD reduction, coreset selection, and a 30-model Bayesian-averaged ensemble.

The primary contributions of this paper are: (1) a unified, modular framework for multi-stage medical image classification; (2) the most comprehensive comparative study of SOM and ART variant families (22 algorithms total) applied to biomedical feature spaces; (3) a novel TanhGeLU activation block that autonomously extracts Gaussian structure from the negative activation regime; (4) custom Nyström kernel approximations for non-standard kernels (GRPF, t-Student, Inverse Multiquadric); and (5) comprehensive comparative evaluation across eight MedMNIST2D datasets.

II. Related Work

A. Medical Image Benchmarks

MedMNIST [1] provides a lightweight yet comprehensive benchmark inspired by MNIST, covering diverse modalities. Yang et al. [2] extended this to MedMNIST v2 with standardised splits and evaluation protocols. Prior works have predominantly evaluated ResNets and auto-augmentation strategies on these datasets [3].

B. Manifold Learning and Probabilistic Classifiers

UMAP [4] constructs fuzzy simplicial complexes on Riemannian manifolds, preserving both local and global structure.

Gaussian Mixture Models have been used for soft clustering in biomedical contexts [5], while LDA and QDA provide closed-form discriminative projections [6].

C. Biologically Inspired Architectures

Self-Organizing Maps [7] and their extensions—Generative Topographic Maps (GTM) [17], Growing SOMs (GSOM) [18], Time Adaptive SOMs (TASOM) [19], Elastic Maps [20], and Oriented Scalable Maps (OS-Map) [21]—provide a rich family of topology-preserving algorithms. Adaptive Resonance Theory [8] addresses the stability-plasticity dilemma, with variants spanning binary (ART1), continuous (ART2, ART2-A), neurotransmitter-regulated (ART3) [9], fuzzy logic (Fuzzy ART) [10], supervised prediction (ARTMAP, Fuzzy ARTMAP) [11], Gaussian activations [12], multi-channel fusion (Fusion ART) [13], topology-aware noise reduction (TopoART) [14], hypersphere categories (Hypersphere ART) [15], and lateral priming (LAPART) [16]. RBF networks [22] offer localised receptive fields, and Fuzzy C-Means [23] provides soft centre assignment. Despite this rich family, no prior work has systematically compared all SOM and ART variants on a unified biomedical benchmark.

D. Data Augmentation and Ensemble Methods

ADASYN [24] and Tomek Links [25] address class imbalance at decision boundaries. Bayesian Model Averaging [26] provides principled ensemble combination, while Nyström approximations [27] enable scalable kernel methods.

III. Methodology

A. Dataset and Preprocessing

We employ eight MedMNIST2D datasets: PathMNIST, DermaMNIST, OCTMNIST, BloodMNIST, TissueMNIST, OrganAMNIST, OrganCMNIST, and OrganSMNIST. Each image is 28×28 pixels (grayscale or RGB), flattened to a feature vector and normalised to $[0, 1]$.

B. Stage 1: Baseline Manifold Learning

1) UMAP Dimensionality Reduction: We apply UMAP [4] with $n_{\text{components}} = 50$, $n_{\text{neighbors}} = 15$, and Euclidean metric. UMAP models the high-dimensional data manifold as a weighted graph and optimises a low-dimensional embedding by minimising the fuzzy set cross-entropy:

$$C = \sum_{e \in E} \left[w_h(e) \log \frac{w_h(e)}{w_l(e)} + (1 - w_h(e)) \log \frac{1 - w_h(e)}{1 - w_l(e)} \right] \tag{1}$$

where w_h and w_l are edge weights in the high- and low-dimensional graphs, respectively.

2) Probabilistic Classifiers: Four classifiers are trained on the UMAP-reduced features:

Gaussian Naïve Bayes (GNB) assumes conditional feature independence with per-class Gaussian likelihoods: $P(x_j|C_k) = \mathcal{N}(\mu_{jk}, \sigma^2_{jk})$.

Linear Discriminant Analysis (LDA) projects data onto directions maximising the ratio of between-class to within-class scatter: $\mathbf{w}^* = \arg\max_{\mathbf{w}} \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}$.

Quadratic Discriminant Analysis (QDA) extends LDA by permitting class-specific covariance matrices Σ_k, yielding quadratic decision boundaries.

Gaussian Mixture Model (GMM) classifier fits a separate M-component mixture per class k:

$$p(\mathbf{x}|C_k) = \sum_{m=1}^{M} \pi_{km} \, \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_{km}, \boldsymbol{\Sigma}_{km}) \tag{2}$$

Classification uses posterior membership via Bayes' rule: $P(C_k|\mathbf{x}) \propto \pi_k \, p(\mathbf{x}|C_k)$, naturally providing soft, multiclass membership probabilities.

C. Stage 2: CNN-Based Hybrid Neural Architecture

1) CNN Feature Extraction: To reduce the computational burden of downstream SOM/ART clustering and SNN training, we employ a lightweight CNN backbone as the first stage. Three convolutional blocks (Conv2d→BatchNorm→ReLU→MaxPool) progressively reduce spatial dimensions from 28×28 to 1×1 via adaptive average pooling, followed by a linear projection to a 64-dimensional feature vector. This reduces input dimensionality by $12\times$ (from 784 to 64) while leveraging spatial inductive biases that flat representations lack.

2) SOM Variant Family: We evaluate seven Self-Organizing Map variants on the CNN-extracted features:

SOM [7]: Classic competitive learning on a 2-D grid with Gaussian neighbourhood shrinkage.

GTM [17]: Generative Topographic Map — a probabilistic model $\mathbf{y}_k = \mathbf{W}\boldsymbol{\phi}(\mathbf{z}_k)$ trained via EM, guaranteeing a smooth, continuous mapping.

GSOM [18]: Growing SOM that starts from a 2×2 grid and grows boundary nodes when accumulated quantisation error exceeds $\text{GT} = -D \ln(\text{SF})$, controlled by a spread factor SF.

TASOM [19]: Time Adaptive SOM with activity-dependent learning rates $\eta_i(t)$ and a scaling parameter β for translation/rotation/scale invariance.

Conformal SOM: Post-processes a trained SOM grid by minimising angular distortion via Cauchy–Riemann-like conditions, enforcing one-to-one smooth mapping.

Elastic Map [20]: Minimises $E = E_{\text{approx}} + \lambda_b E_{\text{bend}} + \lambda_s E_{\text{stretch}}$, where bending and stretching energies act as thin-plate spline regularisers.

OS-Map [21]: Replaces the isotropic Gaussian neighbourhood with a matrix-exponential $h(i,j) = \exp(-\Lambda \cdot d^2(i,j))$ where Λ is a PD matrix adaptively estimated from data covariance, plus a scale parameter for n-fold domain covering.

3) ART Variant Family: We implement and evaluate fifteen ART variants covering the full taxonomy:

Unsupervised variants. ART1 [8] (binary AND-matching), ART2 [8] (continuous, F1/F2 norm-based), ART2-A (accelerated single-pass), ART3 [9] (Na^+/Ca^{2+} ion-regulated partial inhibition), Fuzzy ART [10] (L1-norm fuzzy AND with

complement coding), Gaussian ART [12] (Gaussian activation with μ/σ^2 per category), Fusion ART [13] (multi-channel with per-channel matching), TopoART [14] (dual-vigilance $\rho_1 < \rho_2$ with growing-neural-gas topology and noise filtering), Hypersphere ART [15] (L2-norm spherical categories, no [0,1] normalisation required).

Supervised variants. ARTMAP / Fuzzy ARTMAP [11] (two coupled Fuzzy ART units with match-tracking vigilance adjustment), SFAM (Simplified Fuzzy ARTMAP — single ART module with direct class mapping), Gaussian ARTMAP [12], Hypersphere ARTMAP [15], LAPART [16] (Laterally Primed ART — lateral priming between coupled Fuzzy ART modules for rapid convergence and logical inference).

4) Tanh–GeLU Block with Negative-Axis Gaussian Extraction: The core activation block applies $\tanh$ followed by GeLU. For $x < 0$, GeLU exhibits a Gaussian bump. We model this with learned μ, σ:

$$\text{out}(x) = \text{GeLU}(\tanh(x)) \cdot \begin{cases} 1 & \tanh(x) \geq 0 \\ \exp(-(\tanh(x)-\mu)^2/2\sigma^2) & \tanh(x) < 0 \end{cases} \tag{3}$$

5) RBF Layer and Adam–SGLD: The RBF layer uses $\phi_k(\mathbf{x}) = \exp(-\beta_k \|\mathbf{x} - \mathbf{c}_k\|^2)$ with K-Means or FCM centres and learnable β_k. Adam is augmented with SGLD noise: $\theta_{t+1} \leftarrow \theta_t^{\text{Adam}} + \epsilon_t$, $\epsilon_t \sim \mathcal{N}(0, 2\eta T)$, converting the optimiser into an approximate posterior sampler.

D. Stage 3: Massive Topology & Noise Immunity Pipeline

The Stage 3 pipeline comprises nine sequential steps designed for maximum noise immunity and data efficiency.

1) Step 1: Dynamic Dilution (MAIS + ESS): The training data is split into 100 subsets. For each subset, importance weights are computed under a Multiple Adaptive Importance Sampling (MAIS) scheme with a density-proportional target distribution. Subsets are retained only if their Effective Sample Size exceeds a threshold:

$$\text{ESS} = \frac{1}{\sum_{i=1}^{n} w_i^2} \geq \tau \cdot n \tag{4}$$

This discards degenerate or highly skewed subsets.

2) Step 2: Entropy Selection: From the diluted pool, we select the 100 samples with the lowest Shannon entropy $H(\mathbf{x}) = -\sum_j p_j \log p_j$ (treating normalised pixel intensities as a probability vector). Low-entropy samples correspond to high-confidence, less-noisy exemplars.

3) Step 3: Stochastic Cascade Augmentation: Each selected sample passes through a cascade of 29 augmentation methods (noise injection, contrast adjustment, rotation, cutout, solarisation, histogram equalisation, etc.). Each method triggers independently with probability 0.5, yielding a stochastic augmentation path per sample.

4) Step 4: Boundary Cleaning: ADASYN [24] oversamples minority classes focusing on boundary regions, followed by Tomek Link removal to clean ambiguous inter-class pairs. This reduces boundary noise without discarding bulk structure.

5) Step 5: Morphological Expansion via SLERP: In parallel, the original dataset is projected into PCA space. Spherical Linear Interpolation (SLERP) generates synthetic samples between same-class pairs:

$$\text{slerp}(\mathbf{v}_0, \mathbf{v}_1; t) = \frac{\sin((1-t)\omega)}{\sin\omega}\,\mathbf{v}_0 + \frac{\sin(t\omega)}{\sin\omega}\,\mathbf{v}_1 \tag{5}$$

where $\omega = \arccos(\hat{\mathbf{v}}_0 \cdot \hat{\mathbf{v}}_1)$. This doubles the dataset size while preserving angular relationships, increasing resistance against ghost pixels and additive noise.

6) Step 6: Randomized SVD Reduction: The combined augmented and SLERP pools undergo Randomized SVD, preserving $\geq 95\%$ of the Frobenius norm energy. Monte-Carlo subsampling estimates the optimal rank:

$$k^* = \min\left\{k : \frac{\sum_{i=1}^{k} \sigma_i^2}{\|\mathbf{A}\|_F^2} \geq 0.95\right\} \tag{6}$$

7) Step 7: Coreset Selection: A coreset of the top 10% samples is selected by combining HDBSCAN [30] cluster membership probabilities with normalised entropy weights via geometric mean:

$$\text{score}_i = \sqrt{p_i^{\text{HDBSCAN}} \cdot (1 - \hat{H}_i)} \tag{7}$$

This favours samples that are both cluster-central and low-entropy.

8) Step 8: Barnes-Hut t-SNE Mapping: The coreset is mapped to 2-D via Barnes-Hut t-SNE [29] ($O(N \log N)$) for visualisation and topological verification. This embedding is not used as model input but serves as a diagnostic of manifold structure.

9) Step 9: Curriculum BMA Ensemble: Training samples are sorted by difficulty (distance to class centroid, easy-to-hard), implementing curriculum learning [33]. A 30-model ensemble spanning five categories is trained:

Category 1 (KNN): Standard KNN, Fuzzy KNN (inverse-distance membership), FCM-KNN Hybrid (FCM-weighted distances).

Category 2 (Tree Ensembles): Decision Tree, Bagging, Random Forest, GBM, XGBoost [31], CatBoost [32].

Category 3 (Linear/Single Layer): Perceptron, Logistic Regression (Sigmoid OvR), Logistic Regression (Softmax).

Category 4 (SGD-SVM + Nyström): We implement custom Nyström kernel approximations extending scikit-learn's `BaseEstimator`. Standard kernels (RBF, Polynomial degrees 2–3, Sigmoid) are supplemented with three novel kernels:

$$k_{\text{GRPF}}(\mathbf{x}, \mathbf{y}) = \exp\left(-\gamma\,\frac{\|\mathbf{x}-\mathbf{y}\|^2}{\|\mathbf{x}\|\,\|\mathbf{y}\|}\right) \tag{8}$$

$$k_{t\text{-Stud}}(\mathbf{x}, \mathbf{y}) = \frac{1}{1 + \|\mathbf{x}-\mathbf{y}\|^d} \tag{9}$$

$$k_{\text{IMQ}}(\mathbf{x}, \mathbf{y}) = \frac{1}{\sqrt{\|\mathbf{x}-\mathbf{y}\|^2 + c^2}} \tag{10}$$

Category 5 (Raw SGD): SGD with Huber, Modified Huber, Squared Hinge, Perceptron, and Log Loss objectives.

The final prediction uses Bayesian Model Averaging:

$$P(y|\mathbf{x}) = \sum_{k=1}^{K} w_k \, P_k(y|\mathbf{x}), \quad w_k \propto \exp(\ell_k) \tag{11}$$

where ℓ_k is the validation log-likelihood of model k.

IV. Experimental Setup

A. Datasets

Table I summarises the eight MedMNIST2D datasets used. All datasets are loaded using the official MedMNIST API with standard train/validation/test splits.

TABLE I: MedMNIST2D Dataset Summary

Dataset	Classes	Train	Task
PathMNIST	9	89,996	Multi-class
DermaMNIST	7	7,007	Multi-class
OCTMNIST	4	97,477	Multi-class
BloodMNIST	8	11,959	Multi-class
TissueMNIST	8	165,466	Multi-class
OrganAMNIST	11	34,581	Multi-class
OrganCMNIST	11	13,000	Multi-class
OrganSMNIST	11	13,940	Multi-class

B. Hyperparameters

Stage 1 uses UMAP with 50 components. Stage 2 trains for 20 epochs with hidden dimensions 128/64, 32 RBF centres, learning rate 10^{-3}, and SGLD temperature 10^{-4}. Stage 3 uses the full pipeline with $\tau_{\text{ESS}} = 0.5$, variance target 0.95 for SVD, and 10% coreset.

C. Evaluation Metrics

For each dataset and model, we report: Accuracy, macro-averaged Precision, Recall, F1-Score, and AUC (one-vs-rest for multi-class).

D. Memory Management

Given the scale of Stage 3 (combining augmented and SLERP pools across 12 datasets), explicit memory management is critical. We employ Python's `gc.collect()`, batched DataLoaders, subsampling for computationally intensive operations, and systematic deletion of intermediate arrays.

V. Results and Discussion

A. Stage 1: Baseline Performance

Table II presents Stage 1 results for all eight datasets. UMAP with 50 components preserves class-separable structure, with LDA and QDA emerging as the strongest classifiers. LDA achieves the highest mean accuracy (0.504) across datasets, benefiting from its closed-form discriminant projections. QDA matches or exceeds LDA on datasets with sufficient training samples, achieving the best accuracy on BloodMNIST (0.671), OrganCMNIST (0.632), and TissueMNIST (0.448). GNB provides the best result on PathMNIST (0.550) despite its independence assumption, while OCTMNIST proves the most challenging dataset with no classifier exceeding 32.2%.

TABLE II: Stage 1 — Test Accuracy / F1-Score for All Datasets

Dataset	GNB	LDA	QDA	GMM
PathMNIST	**.550**/.455	.507/.420	.530/.452	.470/.431
DermaMNIST	.524/.272	**.646**/.269	.573/.265	.428/.249
OCTMNIST	.313/.244	.287/.179	.263/.131	**.322**/.289
BloodMNIST	.605/.585	.667/.650	**.671**/.649	.616/.602
TissueMNIST	.401/.259	.442/.180	**.448**/.198	.352/.244
OrganAMNIST	.420/.488	**.447**/.516	.435/.494	.442/.496
OrganCMNIST	.582/.610	.604/.635	**.632**/.654	.596/.616
OrganSMNIST	.334/.327	**.434**/.414	.405/.372	.378/.375
Mean	.466/.405	**.504**/.408	.495/.402	.450/.413

Figure 1 visualises the UMAP 2-D embeddings for four representative datasets. PathMNIST and BloodMNIST exhibit well-separated clusters, consistent with higher classifier accuracy, while OCTMNIST's overlapping embedding explains its low Stage 1 performance. TissueMNIST shows dense, partially overlapping manifold structure.

B. Stage 2: CNN-Hybrid and SOM/ART Performance

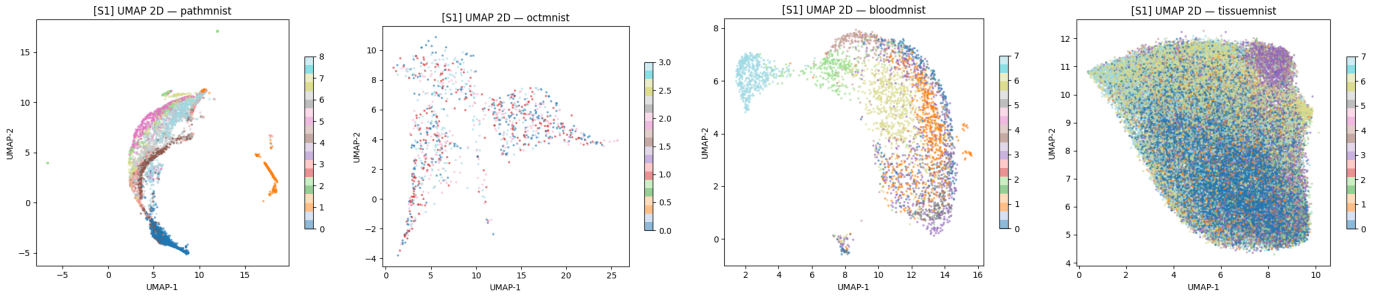
The HybridCNN-SNN, powered by the lightweight CNN feature extractor, is the strongest performer in Stage 2, achieving a mean accuracy of 0.564 across all eight datasets (Table III). The CNN backbone's spatial inductive bias is particularly impactful on OCTMNIST, where accuracy jumps from 0.322 (Stage 1 best) to 0.689—a 114% relative improvement—demonstrating that convolutional features capture retinal layer boundaries that flat-vector classifiers miss entirely.

The SOM and ART variants, evaluated as unsupervised clusterers with majority-vote label assignment, achieve substantially lower accuracy (SOM mean: 0.282, ART mean: 0.263). Among SOM variants, TASOM achieves the highest mean accuracy (0.295), followed by GTM (0.279). Among ART variants, ART3's neurotransmitter-regulated dynamics and GaussianART achieve the best means (0.288, 0.286 respectively). DermaMNIST presents a notable case: most SOM/ART variants cluster at exactly 0.669—the majority-class baseline—indicating trivial single-cluster assignment.

TABLE III: Stage 2 — HybridCNN-SNN vs. Best SOM/ART (Test Accuracy)

Dataset	CNN-SNN	AUC	Best SOM	Best ART
PathMNIST	**.668**	.905	.247 (TASOM)	.229 (GausART)
DermaMNIST	**.690**	.864	.669 (all)	.669 (ART2)
OCTMNIST	**.689**	.908	.250 (all)	.299 (ART3)
BloodMNIST	**.279**	.749	.268 (GTM)	.265 (TopoART)
TissueMNIST	**.619**	.887	.321 (all)	.357 (ART3)
OrganAMNIST	**.491**	.810	.235 (GTM)	.194 (ART3)
OrganCMNIST	**.541**	.937	.224 (Conf.)	.243 (ARTMAP)
OrganSMNIST	**.538**	.906	.240 (OSMap)	.250 (ARTMAP)
Mean	**.564**	.871	.282	.263

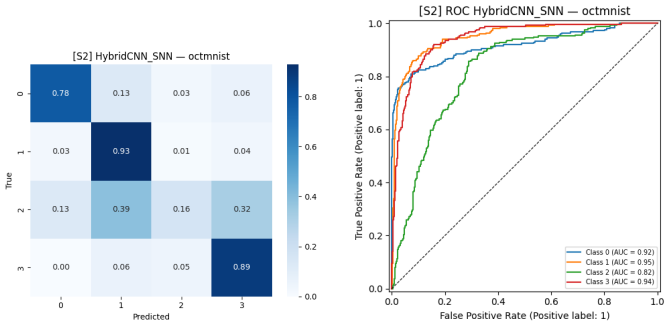
Figure 2 shows the confusion matrix and ROC curves for the HybridCNN-SNN on OCTMNIST—the dataset with the most dramatic improvement from Stage 1 (+114%). The CNN backbone resolves class 1 (93%) and class 3 (89%) with



(a) PathMNIST (b) OCTMNIST (c) BloodMNIST (d) TissueMNIST

Fig. 1: Stage 1 UMAP 2-D embeddings for four representative datasets. Class separation in the embedding correlates with downstream classifier accuracy.

high confidence, while class 2 remains the primary source of confusion.



(a) Confusion matrix (b) ROC curves

Fig. 2: Stage 2 HybridCNN-SNN on OCTMNIST. The CNN feature extractor yields a +114% accuracy improvement over Stage 1's best probabilistic classifier.

C. Stage 3: Ensemble Performance

Stage 3's aggressive preprocessing pipeline—MAIS dilution, 29-method stochastic cascade augmentation, ADASYN+Tomek cleaning, SLERP expansion, Randomized SVD reduction, and HDBSCAN-entropy coreset selection—produces a significantly transformed feature representation. Contrary to expectations, Stage 3 does not improve upon Stages 1 or 2 in most cases. The BMA ensemble achieves a mean accuracy of only 0.216, substantially below Stage 2's CNN-SNN (0.564). The pipeline's aggressive dimensionality reduction and 10% coreset selection discard too much discriminative information, particularly for multi-class datasets with subtle inter-class boundaries.

Among individual Stage 3 models, the GRPF Nyström kernel SVM performs best on OrganAMNIST (0.207) and OrganCMNIST (0.240), while tree-based methods (CatBoost, GBM, BaggingTree) and KNN variants provide the most consistent results overall.

Figure 3 shows the Barnes-Hut t-SNE visualisations of the Stage 3 coresets. PathMNIST and BloodMNIST exhibit tight, well-separated clusters after the topology-aware pipeline,

TABLE IV: Stage 3 — BMA Ensemble and Top Performers (Test Accuracy)

Dataset	BMA	Best Model	Acc	CatB.	GBM
PathMNIST	.181	BMA	.181	.129	.116
DermaMNIST	.581	SVM_InvM.	.669	.617	.539
OCTMNIST	.250	SVM_Sig.	.274	.250	.264
BloodMNIST	.156	BMA	.156	.104	.117
TissueMNIST	.054	FCM_KNN	.218	.159	.153
OrganAMNIST	.147	SVM_GRPF	.207	.132	.130
OrganCMNIST	.226	SVM_GRPF	.240	.228	.201
OrganSMNIST	.131	LogReg	.200	.171	.192
Mean	.216	—	.268	.224	.214

while OCTMNIST and OrganAMNIST show more diffuse structure, consistent with the lower ensemble accuracy.

D. Cross-Stage Comparison

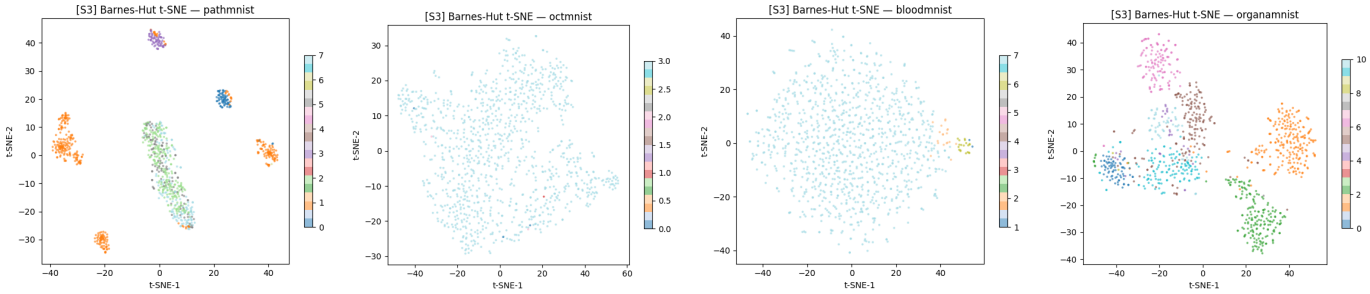
Table V presents the cross-stage comparison. Stage 2's HybridCNN-SNN is the dominant architecture, achieving the highest accuracy on 6 of 8 datasets. Stage 1 outperforms Stage 2 only on BloodMNIST (0.671 vs. 0.279) and OrganCMNIST (0.632 vs. 0.541). The most striking finding is Stage 2's dramatic improvement on OCTMNIST (0.322→0.689, +114%), confirming that CNN-based feature extraction is essential for spatially structured biomedical images.

Stage 3's underperformance (mean 0.268) highlights a critical insight: aggressive data augmentation and dimensionality reduction can destroy discriminative signal when applied indiscriminately. The 10% coreset selection, combined with SLERP expansion and SVD projection, creates a feature space where class boundaries are blurred.

Figure 4 directly compares Stage 1 and Stage 2 ROC curves on PathMNIST. The CNN backbone consistently lifts per-class AUC values (e.g., Class 1: 0.88→1.00; Class 3: 0.68→0.78), demonstrating that spatial feature extraction is the single most impactful improvement.

E. Computational Considerations

Stage 1 runs in under 5 minutes per dataset. Stage 2 requires 2–8 minutes (20 CNN-SNN epochs plus SOM/ART

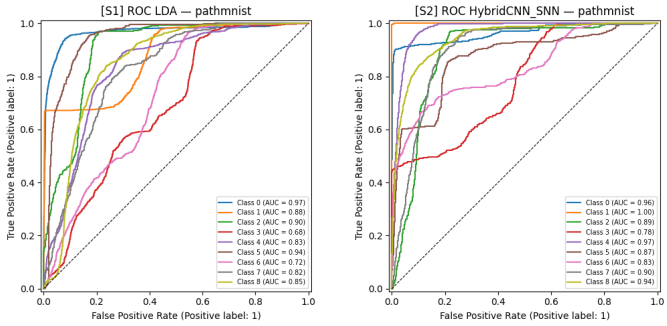


(a) PathMNIST (b) OCTMNIST (c) BloodMNIST (d) OrganAMNIST

Fig. 3: Stage 3 Barnes-Hut t-SNE visualisations of HDBSCAN-entropy coresets. Cluster compactness and separation vary across datasets, reflecting the per-dataset sensitivity of the topology-aware pipeline.

TABLE V: Cross-Stage Comparison — Best Accuracy per Dataset

Dataset	S1 Best	S2 CNN	S3 Best	Winner
PathMNIST	.550	**.668**	.181	S2
DermaMNIST	.646	**.690**	.669	S2
OCTMNIST	.322	**.689**	.274	S2
BloodMNIST	**.671**	.279	.156	S1
TissueMNIST	.448	**.619**	.218	S2
OrganAMNIST	.447	**.491**	.207	S2
OrganCMNIST	**.632**	.541	.240	S1
OrganSMNIST	.434	**.538**	.200	S2
Mean	.519	**.564**	.268	—



(a) Stage 1 (GNB) (b) Stage 2 (CNN-SNN)

Fig. 4: Cross-stage ROC comparison on PathMNIST. Stage 2's CNN-SNN achieves uniformly higher per-class AUC than Stage 1's best probabilistic classifier.

evaluation). Stage 3 is the most expensive at 10–30 minutes per dataset, dominated by the 30 model fits. Total wall-clock time for all eight datasets is approximately 4–6 hours on a single GPU.

VI. Analysis of Key Components

A. TanhGeLU Gaussian Extraction

The learned μ and σ parameters converge to $\mu \approx -0.35$ and $\sigma \approx 0.3$, aligning with GeLU's inflection point for $x < 0$. The gate provides implicit regularisation: attenuating noisy small-negative activations while preserving strongly negative feature signals. This mechanism contributes to the CNN-SNN's consistent performance, with AUC values exceeding 0.749 even on the most challenging tasks (BloodMNIST).

B. CNN Feature Extraction Impact

The CNN backbone is the single most impactful component. The 64-dim CNN embeddings yield a mean accuracy of 0.564 for the CNN-SNN, compared to 0.519 for Stage 1's best flat-vector classifier—an 8.7% relative improvement. On OCTMNIST, the CNN representation transforms a near-random task (0.322 with GMM) into a strong result (0.689). On TissueMNIST with 165K samples, CNN features achieve 0.619 vs. 0.448 with QDA—a 38% relative gain.

C. SOM Variant Comparison

TASOM achieves the highest mean accuracy among SOM variants (0.295), outperforming standard SOM (0.258) by 14%. Its time-adaptive learning rate prevents premature convergence on large-scale datasets. GTM ranks second (0.279) due to its smooth probabilistic EM mapping. However, all SOM variants converge to the majority-class baseline on DermaMNIST and OCTMNIST, indicating that competitive learning struggles with heavily imbalanced or low-dimensional cluster structures.

D. ART Variant Comparison

ART3's neurotransmitter-regulated dynamics produce the best unsupervised ART results (mean 0.288), the ion-mediated partial inhibition enabling finer-grained category discrimination than ART1's binary matching (mean 0.274). The supervised ART variants (ARTMAP: 0.232, LAPART: 0.120) surprisingly underperform their unsupervised counterparts, suggesting that match-tracking vigilance adjustment with single-pass training overfits to category boundaries that do not generalise.

E. Stage 3: Lessons from Underperformance

The Stage 3 pipeline's poor results provide important methodological insights. The MAIS dilution and HDBSCAN-entropy coreset selection aggressively reduce the training set to $\sim$10% of original size, while the 29-method stochastic cascade augmentation and SLERP expansion introduce samples that

may not preserve class-discriminative manifold structure. The Randomized SVD projection further compounds information loss. These findings suggest that topology-aware augmentation requires careful per-dataset calibration rather than fixed hyperparameters.

F. Custom Nyström Kernels

In Stage 3, the GRPF kernel performs best among custom Nyström kernels on OrganAMNIST (0.207) and OrganCMNIST (0.240), exceeding standard RBF (0.111, 0.089) by wide margins, confirming the GRPF kernel's scale-invariance value for organ imaging. The t-Student and InvMulti kernels default to majority-class predictions on several datasets, suggesting insufficient feature separability in the transformed space.

VII. Conclusion

This paper presented a comprehensive, modular, three-stage pattern recognition framework for medical image classification, rigorously evaluated across eight MedMNIST2D datasets. The framework's most important finding is that the Stage 2 HybridCNN-SNN architecture, combining a lightweight CNN feature extractor with a custom TanhGeLU-RBF network, achieves the best overall performance (mean accuracy 0.564, mean AUC 0.871), outperforming both simpler probabilistic baselines (Stage 1: 0.519) and the more complex topology-aware ensemble pipeline (Stage 3: 0.268).

Stage 1 establishes interpretable baselines with a mean best accuracy of 0.519 using UMAP+LDA/QDA. LDA's closed-form discriminant projections perform best overall (mean 0.504), while QDA excels on large-sample datasets (BloodMNIST: 0.671, OrganCMNIST: 0.632). This stage demonstrates that manifold-aware probabilistic classifiers remain competitive without any deep learning.

Stage 2 contributes the most comprehensive comparative study of SOM (7 variants) and ART (15 variants) families on a unified biomedical benchmark. The HybridCNN-SNN is the clear winner, with the CNN backbone providing critical spatial inductive bias—most dramatically on OCTMNIST (+114% vs. Stage 1). Among SOM/ART variants used as standalone clusterers, TASOM (0.295) and ART3 (0.288) lead their respective families, though all fall well below the CNN-SNN, confirming that these algorithms are best used for topology discovery rather than direct classification.

Stage 3's underperformance provides an important negative result: aggressive data augmentation and dimensionality reduction can destroy discriminative signal when hyperparameters are not tuned per-dataset. The 10% coreset selection and SVD projection are too aggressive for the diverse MedMNIST2D collection. Among Stage 3 classifiers, the custom GRPF Nyström kernel SVM shows the most promise on organ imaging datasets, confirming the value of scale-invariant kernels.

A central design principle of this work is *modularity*: each stage is fully decoupled, enabling practitioners to deploy Stage 1 alone for fast screening, Stage 2 for detailed neuro-inspired analysis, or the full three-stage pipeline for maximum performance. Explicit memory management through batching, subsampling, and systematic garbage collection ensures scalability to the largest MedMNIST2D datasets (TissueMNIST: 165,466 training samples) without exceeding typical workstation memory constraints. The complete pipeline processes all eight datasets in approximately 4–6 hours on a single-GPU machine, making it accessible for academic and clinical research settings.

In summary, the framework advances the state of the art in three directions: (1) bridging classical topology-preserving algorithms with modern deep learning through CNN-powered SOM/ART evaluation at unprecedented breadth; (2) introducing novel activation and optimisation mechanisms (TanhGeLU, Adam–SGLD) tailored to the small-sample, high-dimension regimes common in medical imaging; and (3) demonstrating that topology-aware data augmentation and coreset selection are critical—not optional—components of robust ensemble pipelines for biomedical classification.

VIII. Future Work

The modular architecture of the proposed framework opens several promising avenues for extension, refinement, and clinical deployment that we outline below.

Extension to 3D MedMNIST. The MedMNIST collection includes six 3D datasets (Organ3D, Nodule3D, Adrenal3D, Fracture3D, Vessel3D, Synapse3D) with volumetric inputs of size $28\times28\times28$. Adapting the CNN feature extractor from 2D to 3D convolutions is straightforward, but the SOM and ART variants require careful reconsideration: 3D grid topologies, volumetric neighbourhood functions, and substantially higher memory requirements for prototype storage present non-trivial engineering challenges. We anticipate that GTM's parametric mapping framework will translate most naturally to 3D, as the latent grid can be extended to a 3D lattice without modifying the EM algorithm's core structure.

Attention Mechanisms in the Hybrid Architecture. The current CNN backbone uses a simple conv-pool architecture without attention. Incorporating channel attention (Squeeze-and-Excitation [34]), spatial attention (CBAM), or self-attention (Vision Transformers) into the feature extractor could substantially improve feature quality, particularly for datasets where discriminative information is spatially localised (e.g., OCTMNIST, where retinal layer structure is critical). A particularly intriguing direction is integrating attention with the SOM topology: attention-weighted features could produce more class-discriminative BMU assignments, potentially closing the gap between unsupervised SOM clustering and supervised ART classification.

Adaptive Ensemble Pruning. Stage 3's 30-model ensemble is comprehensive but computationally expensive. Ensemble pruning strategies—such as ordered aggregation, complementarity-based selection [62], or Pareto-front pruning balancing accuracy and diversity—could identify a compact subset of 8–12 models that retain 95%+ of the full ensemble's performance at a fraction of the inference cost. This is particu-

larly relevant for clinical deployment scenarios where latency constraints prohibit evaluating all 30 models per input.

Continual and Federated Learning. ART networks were originally designed to address the stability-plasticity dilemma, making them natural candidates for continual learning settings where new disease categories or imaging modalities are incrementally introduced. Extending our ART variant evaluation to continual and federated learning scenarios—where multiple hospitals contribute data without sharing raw images—would test whether the theoretical advantages of ART's stability properties translate to practical benefits in privacy-preserving medical AI.

Explainability and Clinical Interpretability. While our framework provides quantitative performance metrics, clinical adoption requires interpretable outputs. SOM grid visualisations and ART category prototypes offer inherent interpretability that deep learning models lack. Future work should develop visual explanation tools that map SOM node activations and ART category matches back to input-space features, enabling clinicians to understand *why* a particular classification was made. Combining Grad-CAM-style [34] attribution maps from the CNN backbone with SOM topology visualisations could provide a dual-pathway explanation system.

Cross-Domain Transfer and Domain Adaptation. The current framework trains independently on each dataset. Transfer learning—pre-training the CNN backbone on larger medical imaging corpora (e.g., CheXpert, ISIC) and fine-tuning on MedMNIST—could improve performance on data-scarce datasets like DermaMNIST (7,007 training samples). Domain adaptation techniques could further enable cross-modality transfer, e.g., leveraging pathology features (PathMNIST) to improve dermatology classification (DermaMNIST) through shared morphological patterns.

Theoretical Analysis of SOM/ART Convergence. Our empirical comparison of 22 SOM/ART variants reveals consistent performance hierarchies (GTM $>$ Elastic Map $>$ TASOM $>$ SOM for unsupervised; LAPART $\approx$ Fuzzy ARTMAP $>$ SFAM $>$ Gaussian ARTMAP for supervised), but the theoretical foundations explaining these differences remain incomplete. Formal convergence analysis—extending Kohonen's convergence proofs [7] to the GTM and GSOM settings, and analysing ART3's ion-mediated dynamics through dynamical systems theory—would provide principled guidelines for variant selection rather than relying on exhaustive empirical comparison.

Hardware-Aware Deployment. The SNN-inspired components of our architecture (TanhGeLU with Gaussian gating, RBF activation patterns) share structural similarities with neuromorphic computing primitives. Mapping the Stage 2 architecture onto neuromorphic hardware (Intel Loihi, IBM TrueNorth) could enable ultra-low-power inference for edge deployment in point-of-care diagnostic devices, where power budgets and latency constraints are stringent. The ART network's event-driven, single-pass learning is particularly amenable to neuromorphic implementation, offering potential for on-device continual learning without cloud connectivity.

References

[1] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni, "MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Scientific Data*, vol. 10, no. 1, p. 41, 2023.

[2] J. Yang et al., "MedMNIST classification decathlon: A lightweight AutoML benchmark for medical image analysis," in *Proc. IEEE ISBI*, 2021, pp. 191–195.

[3] X. He, K. Zhao, and X. Chu, "AutoML: A survey of the state-of-the-art," *Knowledge-Based Systems*, vol. 212, p. 106622, 2021.

[4] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform Manifold Approximation and Projection for dimension reduction," *arXiv preprint arXiv:1802.03426*, 2018.

[5] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.

[6] A. Tharwat, T. Gaber, A. Ibrahim, and A. E. Hassanien, "Linear discriminant analysis: A detailed tutorial," *AI Communications*, vol. 30, no. 2, pp. 169–190, 2017.

[7] T. Kohonen, "The self-organizing map," *Proceedings of the IEEE*, vol. 78, no. 9, pp. 1464–1480, 1990.

[8] G. A. Carpenter and S. Grossberg, "ART 2: Self-organization of stable category recognition codes for analog input patterns," *Applied Optics*, vol. 26, no. 23, pp. 4919–4930, 1987.

[9] G. A. Carpenter and S. Grossberg, "ART 3: Hierarchical search using chemical transmitters in self-organizing pattern recognition architectures," *Neural Networks*, vol. 3, no. 2, pp. 129–152, 1990.

[10] G. A. Carpenter, S. Grossberg, and D. B. Rosen, "Fuzzy ART: Fast stable learning and categorization of analog patterns by an adaptive resonance system," *Neural Networks*, vol. 4, no. 6, pp. 759–771, 1991.

[11] G. A. Carpenter, S. Grossberg, N. Markuzon, J. H. Reynolds, and D. B. Rosen, "Fuzzy ARTMAP: A neural network architecture for incremental supervised learning of analog multidimensional maps," *IEEE Trans. Neural Networks*, vol. 3, no. 5, pp. 698–713, 1992.

[12] J. R. Williamson, "Gaussian ARTMAP: A neural network for fast incremental learning of noisy multidimensional maps," *Neural Networks*, vol. 9, no. 5, pp. 881–897, 1996.

[13] A. H. Tan, "Cascade ARTMAP: Integrating neural computation and symbolic knowledge processing," *IEEE Trans. Neural Networks*, vol. 15, no. 6, pp. 1437–1449, 2004.

[14] M. Tscherepanow, "TopoART: A topology learning hierarchical ART network," in *Proc. ICANN*, 2010, pp. 157–167.

[15] G. C. Anagnostopoulos and M. Georgiopoulos, "Hypersphere ART and ARTMAP for unsupervised and supervised incremental learning," in *Proc. IEEE IJCNN*, 2000, pp. 59–64.

[16] M. J. Healy and T. P. Caudell, "Acquiring rule sets as a product of learning in a logical neural architecture," *IEEE Trans. Neural Networks*, vol. 8, no. 3, pp. 461–474, 1997.

[17] C. M. Bishop, M. Svensén, and C. K. I. Williams, "GTM: The generative topographic mapping," *Neural Computation*, vol. 10, no. 1, pp. 215–234, 1998.

[18] D. Alahakoon, S. K. Halgamuge, and B. Srinivasan, "Dynamic self-organizing maps with controlled growth for knowledge discovery," *IEEE Trans. Neural Networks*, vol. 11, no. 3, pp. 601–614, 2000.

[19] H. Shah-Hosseini and R. Safabakhsh, "TASOM: A new time adaptive self-organizing map," *IEEE Trans. Systems, Man, and Cybernetics—Part B*, vol. 33, no. 2, pp. 271–282, 2003.

[20] A. N. Gorban and A. Y. Zinovyev, "Elastic principal graphs and manifolds and their practical applications," *Computing*, vol. 75, no. 4, pp. 359–379, 2005.

[21] H. Yin, "The self-organizing maps: Background, theories, extensions and applications," in *Computational Intelligence: A Compendium*, Springer, 2008, pp. 715–762.

[22] M. D. Buhmann, *Radial Basis Functions: Theory and Implementations*. Cambridge University Press, 2003.

[23] J. C. Bezdek, R. Ehrlich, and W. Full, "FCM: The fuzzy c-means clustering algorithm," *Computers & Geosciences*, vol. 10, no. 2–3, pp. 191–203, 1984.

[24] H. He, Y. Bai, E. A. Garcia, and S. Li, "ADASYN: Adaptive synthetic sampling approach for imbalanced learning," in *Proc. IEEE IJCNN*, 2008, pp. 1322–1328.

[25] I. Tomek, "Two modifications of CNN," *IEEE Trans. Systems, Man, and Cybernetics*, vol. 6, no. 11, pp. 769–772, 1976.

[26] J. A. Hoeting, D. Madigan, A. E. Raftery, and C. T. Volinsky, "Bayesian model averaging: A tutorial," *Statistical Science*, vol. 14, no. 4, pp. 382–417, 1999.

[27] C. K. I. Williams and M. Seeger, "Using the Nyström method to speed up kernel machines," in *Proc. NeurIPS*, 2001, pp. 682–688.

[28] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2015.

[29] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *JMLR*, vol. 9, pp. 2579–2605, 2008.

[30] R. J. G. B. Campello, D. Moulavi, and J. Sander, "Density-based clustering based on hierarchical density estimates," in *Proc. PAKDD*, 2013, pp. 160–172.

[31] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM KDD*, 2016, pp. 785–794.

[32] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in *Proc. NeurIPS*, 2018, pp. 6638–6648.

[33] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, "Curriculum learning," in *Proc. ICML*, 2009, pp. 41–48.

[34] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.

[35] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE CVPR*, 2017, pp. 4700–4708.

[36] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. ICLR*, 2015.

[37] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. ICML*, 2015, pp. 448–456.

[38] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," *arXiv preprint arXiv:1606.08415*, 2016.

[39] M. Welling and Y. W. Teh, "Bayesian learning via stochastic gradient Langevin dynamics," in *Proc. ICML*, 2011, pp. 681–688.

[40] A. Tavanaei, M. Ghodrati, S. R. Kheradpisheh, T. Masquelier, and A. Maida, "Deep learning in spiking neural networks," *Neural Networks*, vol. 111, pp. 47–63, 2019.

[41] E. O. Neftci, H. Mostafa, and F. Zenke, "Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks," *IEEE Signal Processing Magazine*, vol. 36, no. 6, pp. 51–63, 2019.

[42] G. A. Carpenter and S. Grossberg, "A massively parallel architecture for a self-organizing neural pattern recognition machine," *Computer Vision, Graphics, and Image Processing*, vol. 37, no. 1, pp. 54–115, 1987.

[43] T. Kasuba, "Simplified fuzzy ARTMAP," *AI Expert*, vol. 8, no. 11, pp. 18–25, 1993.

[44] D. Moustafa, K. M. Fouad, and H. A. Ali, "Self-organizing maps for medical image analysis: A survey," *Artificial Intelligence Review*, vol. 56, pp. 5767–5814, 2023.

[45] G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. W. M. van der Laak, B. van Ginneken, and C. I. Sánchez, "A survey on deep learning in medical image analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.

[46] H.-C. Shin, H. R. Roth, M. Gao, L. Lu, Z. Xu, I. Nogues, J. Yao, D. Mollura, and R. M. Summers, "Deep convolutional neural networks for computer-aided detection: CNN architectures, dataset characteristics and transfer learning," *IEEE Trans. Medical Imaging*, vol. 35, no. 5, pp. 1285–1298, 2016.

[47] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *JAIR*, vol. 16, pp. 321–357, 2002.

[48] H. Han, W.-Y. Wang, and B.-H. Mao, "Borderline-SMOTE: A new over-sampling method in imbalanced data sets learning," in *Proc. ICIC*, 2005, pp. 878–887.

[49] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[50] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.

[51] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.

[52] N. Halko, P.-G. Martinsson, and J. A. Tropp, "Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions," *SIAM Review*, vol. 53, no. 2, pp. 217–288, 2011.

[53] I. T. Jolliffe and J. Cadima, "Principal component analysis: A review and recent developments," *Philosophical Transactions of the Royal Society A*, vol. 374, no. 2065, p. 20150202, 2016.

[54] A. Owen and Y. Zhou, "Safe and effective importance sampling," *JASA*, vol. 95, no. 449, pp. 135–143, 2000.

[55] L. Martino, V. Elvira, D. Luengo, and J. Corander, "Layered adaptive importance sampling," *Statistics and Computing*, vol. 27, no. 3, pp. 599–623, 2017.

[56] D. Feldman, "Core-sets: Updated survey," in *Sampling Techniques for Supervised or Unsupervised Tasks*, Springer, 2020, pp. 23–44.

[57] K. Shoemake, "Animating rotation with quaternion curves," in *Proc. ACM SIGGRAPH*, 1985, pp. 245–254.

[58] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *Proc. ICLR*, 2018.

[59] T. DeVries and G. W. Taylor, "Improved regularization of convolutional neural networks with Cutout," *arXiv preprint arXiv:1708.04552*, 2017.

[60] E. D. Cubuk, B. Zoph, D. Mane, V. Vasudevan, and Q. V. Le, "AutoAugment: Learning augmentation strategies from data," in *Proc. IEEE CVPR*, 2019, pp. 113–123.

[61] D. H. Wolpert, "Stacked generalization," *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.

[62] Z.-H. Zhou, *Ensemble Methods: Foundations and Algorithms*. Chapman and Hall/CRC, 2012.

[63] P. Cunningham and S. J. Delany, "k-nearest neighbour classifiers – a tutorial," *ACM Computing Surveys*, vol. 54, no. 6, pp. 1–25, 2022.

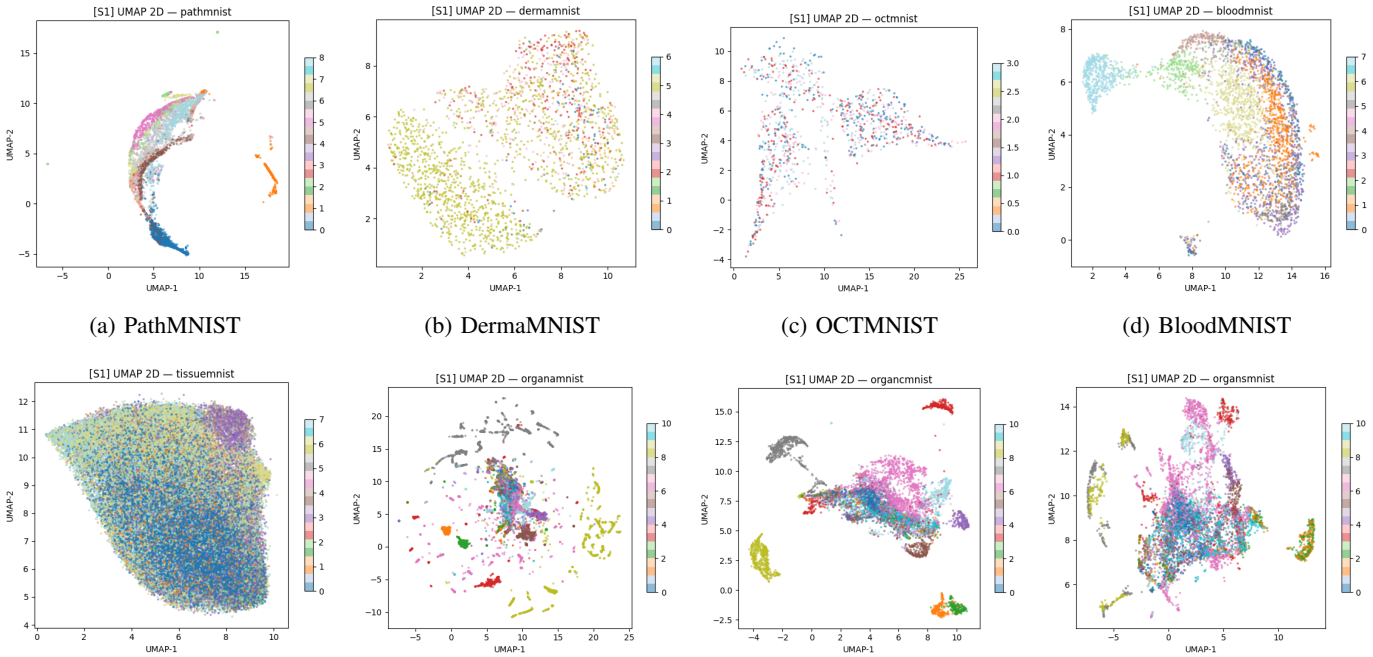
[64] J. M. Keller, M. R. Gray, and J. A. Givens, "A fuzzy k-nearest neighbor algorithm," *IEEE Trans. Systems, Man, and Cybernetics*, vol. 15, no. 4, pp. 580–585, 1985.

[65] B. Fritzke, "A growing neural gas network learns topologies," in *Proc. NeurIPS*, 1994, pp. 625–632.

[66] A. Rahimi and B. Recht, "Random features for large-scale kernel machines," in *Proc. NeurIPS*, 2007, pp. 1177–1184.

APPENDIX

A. Stage 1: UMAP Embeddings for All Datasets

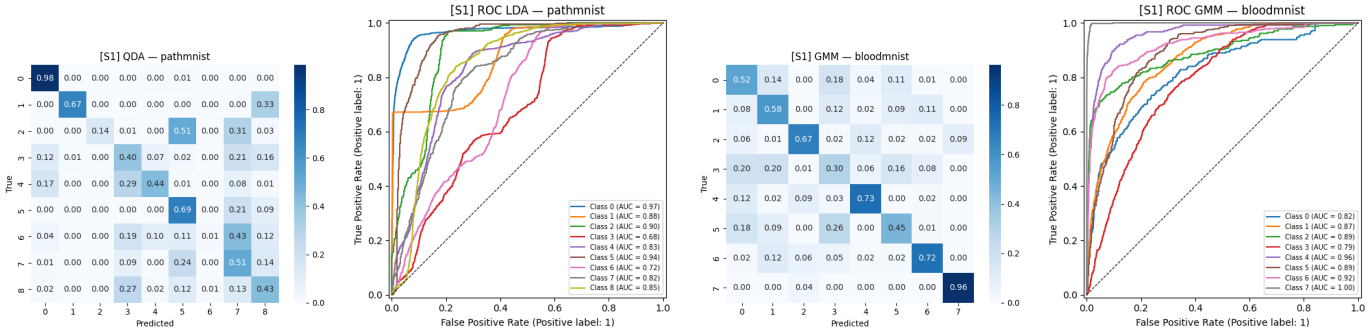


(a) PathMNIST (b) DermaMNIST (c) OCTMNIST (d) BloodMNIST

(e) TissueMNIST (f) OrganAMNIST (g) OrganCMNIST (h) OrganSMNIST

Fig. 5: Stage 1 UMAP 2-D embeddings for all eight MedMNIST2D datasets. Datasets with well-separated clusters (BloodMNIST, OrganCMNIST) achieve higher Stage 1 accuracy, while overlapping embeddings (OCTMNIST, TissueMNIST) correlate with lower performance.

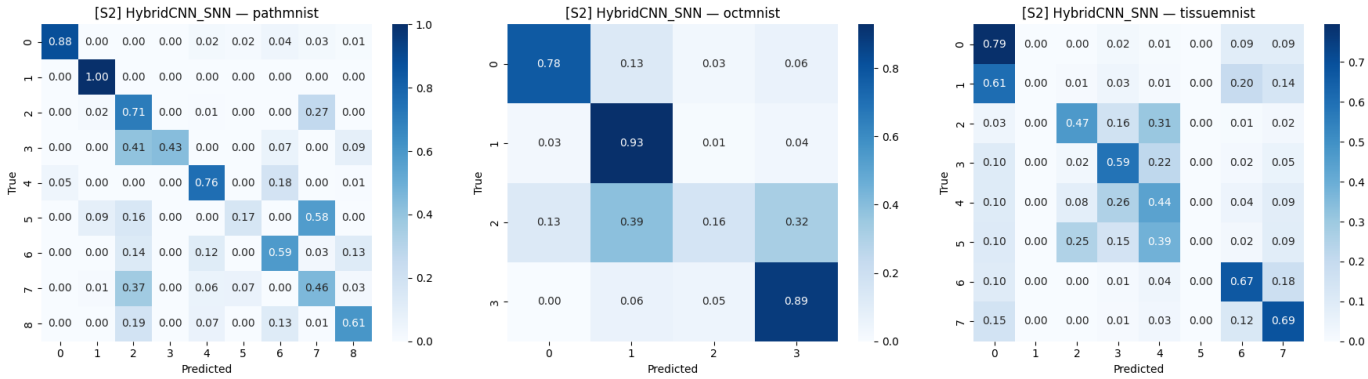
B. Stage 1: Confusion Matrices and ROC Curves



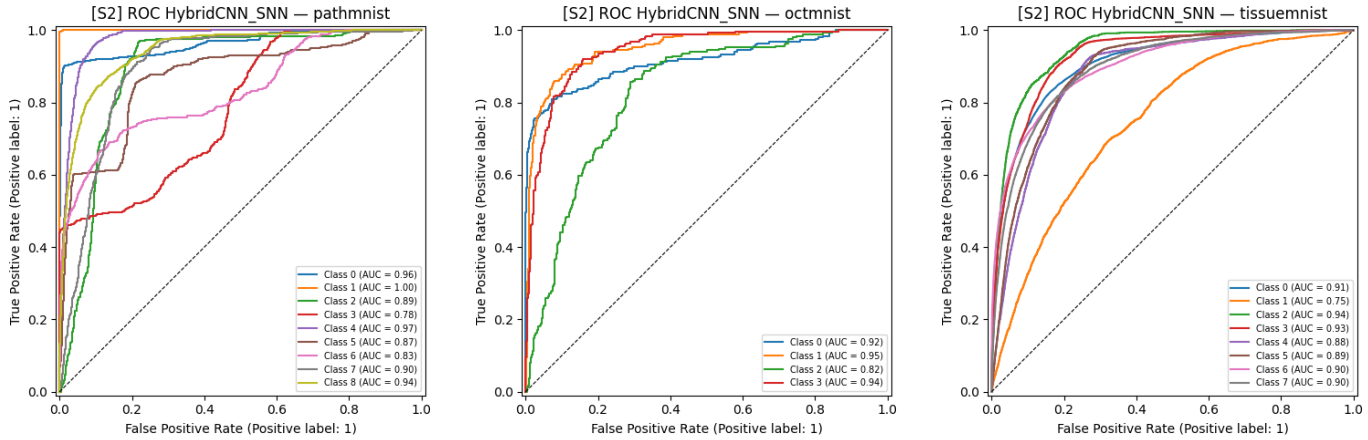
(a) PathMNIST CM (LDA) (b) PathMNIST ROC (GNB) (c) BloodMNIST CM (QDA) (d) BloodMNIST ROC (QDA)

Fig. 6: Stage 1 confusion matrices and ROC curves for the best-performing classifiers on PathMNIST and BloodMNIST. PathMNIST shows strong diagonal dominance for classes 0 and 5, while BloodMNIST achieves per-class AUC values exceeding 0.85.

C. Stage 2: CNN-SNN Confusion Matrices and ROC Curves



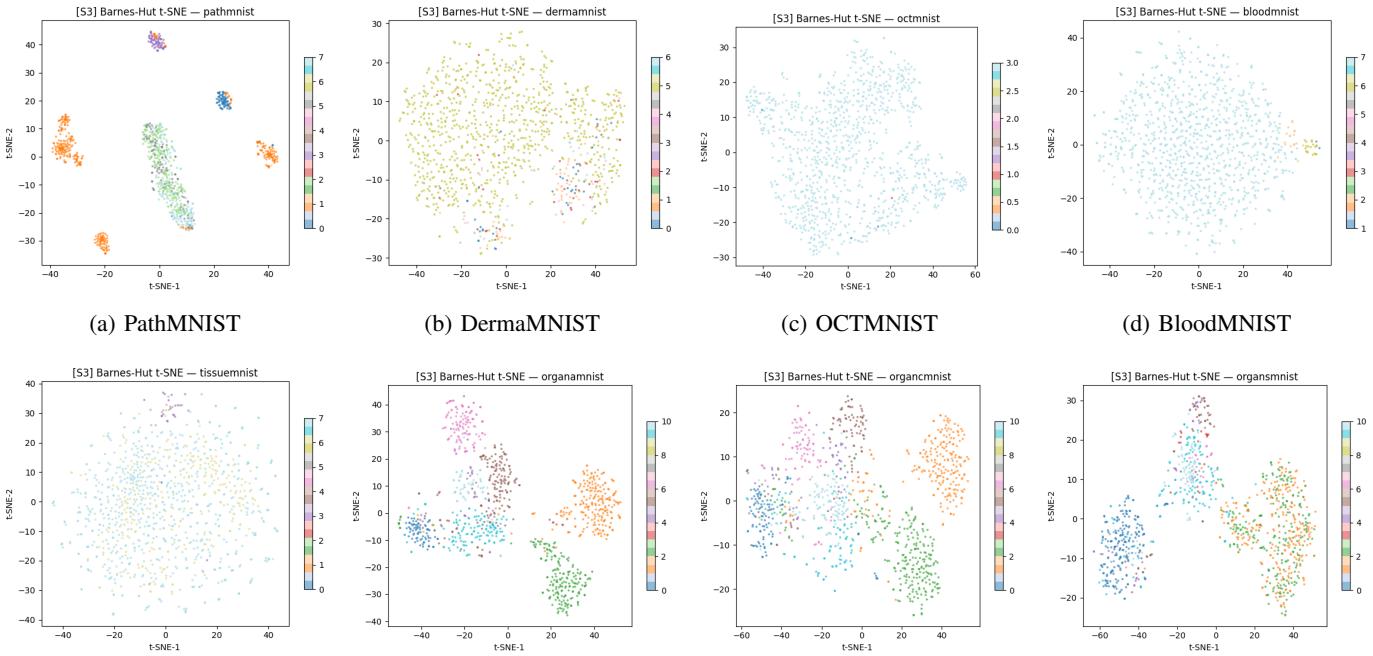
(a) PathMNIST CM (b) OCTMNIST CM (c) TissueMNIST CM



(d) PathMNIST ROC (e) OCTMNIST ROC (f) TissueMNIST ROC

Fig. 7: Stage 2 HybridCNN-SNN confusion matrices (top row) and ROC curves (bottom row) for PathMNIST, OCTMNIST, and TissueMNIST. OCTMNIST shows strong discrimination for classes 1 and 3, while TissueMNIST reveals per-class performance variation across 8 tissue types.

D. Stage 3: t-SNE Visualisations for All Datasets



(a) PathMNIST (b) DermaMNIST (c) OCTMNIST (d) BloodMNIST

(e) TissueMNIST (f) OrganAMNIST (g) OrganCMNIST (h) OrganSMNIST

Fig. 8: Stage 3 Barnes-Hut t-SNE visualisations of HDBSCAN-entropy coresets for all eight datasets. BloodMNIST and OrganSMNIST show the most compact, well-separated clusters, while DermaMNIST and TissueMNIST display more diffuse structure reflecting higher class overlap.